Model Explainer: Medical Equipment Failure Risk Prediction

Week 9 Deliverable | Valentine Nyankwoga

The Analogy

Think of this model as an experienced fleet mechanic who does a quick walk-around inspection of every vehicle each morning before anyone drives off. The mechanic doesn't tear an engine apart to check it; instead they quickly note things like mileage, how long it's been since the last service, and whether anything is vibrating or running hotter than it should. From that quick check, they point at the two or three vehicles that need a closer look today, out of the whole fleet.

That is exactly what this model does for hospital equipment. It cannot diagnose the exact fault, and it isn't a substitute for a hands-on biomedical inspection but every day it scans every dialysis machine, ventilator, X-ray unit, infusion pump, anaesthesia machine and ECG monitor, and quietly says: "start with these, they're the ones most likely to break down in the next 30 days." With a small maintenance team and dozens of machines across a facility, that prioritisation is the whole value of the tool.

The Drivers

The chart below ranks which operational factors the model relies on most heavily when it decides a machine is high-risk. The top three, in plain English:

#1 Days since last maintenance – The longer it's been since a machine was serviced, the more the model treats it as "overdue and unpredictable" just like a car well past its scheduled service.

#2 Vibration level – Unusual shaking or mechanical noise is often the earliest physical sign that moving parts are wearing out, well before a machine actually stops working.

#3 Weekly usage hours – Machines that are run harder and longer accumulate wear faster, a unit in near-constant use is simply exposed to more failure opportunities than one used lightly.

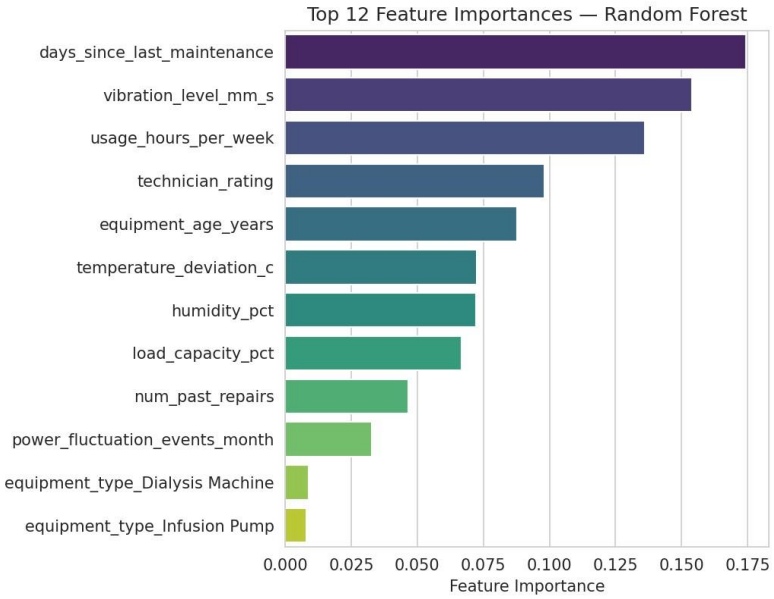


Figure 1. Top feature importances from the Random Forest risk model (held-out test evaluation: Precision 0.29, Recall 0.27, F1 0.28, ROC-AUC 0.70).

Beyond the top three, maintenance history (number of past repairs) and technician service quality also matter: machines with a history of repeat repairs, or whose last service was rated poorly, carry more risk reflecting that recent repair and service quality are almost as predictive as raw usage or age.

Trust & Limits

What the errors mean. On held-out test data, the model correctly identifies roughly one in four genuinely high-risk machines it's shown (Recall ≈ 0.27), and about 29% of its "High Risk" flags turn out to be real. In practice that means:

- **False negatives** (a machine that will fail, but isn't flagged) are the more dangerous error here a missed dialysis machine or ventilator failure is a patient-safety issue, not just a cost. This is why the model was deliberately tuned toward catching more true risk (via class-weighting and SMOTE) rather than optimizing for overall accuracy.
- **False positives** (a flagged machine that turns out to be fine) still cost engineer time and, if frequent, can cause "alert fatigue" where staff start ignoring the tool altogether. The current precision (≈29%) means roughly 2 in 3 flags will not turn into a real fault that trade-off should be tuned against the team's actual inspection capacity as real data comes in.
- **The model is a synthetic-data prototype.** It was trained on simulated equipment logs built to reflect realistic relationships (age, usage, vibration, maintenance gaps), not yet on real Vital Asset sensor/maintenance data. Absolute performance numbers will change once real hospital data is used; the pipeline and evaluation approach are what should carry over.

How operations should use it. Treat every "High Risk" flag as a prompt to inspect sooner, not as a confirmed diagnosis and not as permission to skip inspecting equipment the model didn't flag. Each flag should always be shown alongside its top SHAP drivers (see Section 6 of the accompanying notebook) so the engineer arrives already knowing what to check first vibration, an overdue service, or unusually heavy recent use. The tool is a triage assistant for a limited maintenance team, not a replacement for biomedical engineering judgement.